

ECEN 3730: Lab #2 Report

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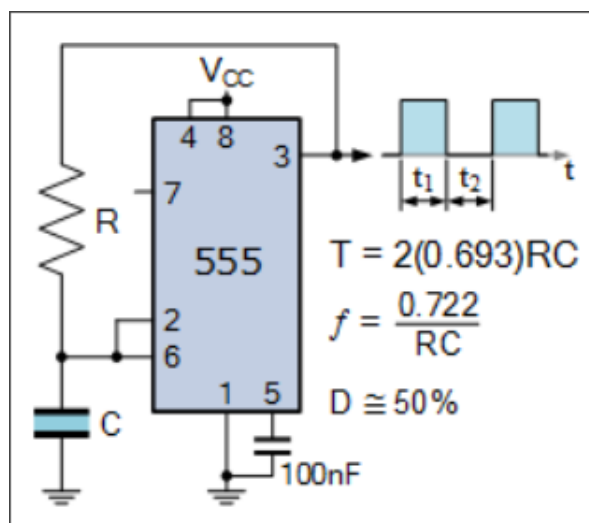
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Overview

In this lab, we begin to focus on the process of developing a PCB from a "back-of-napkin" sketch, focusing on prototyping and verifying your original design with real hardware.

Towards this end, we are gearing up to develop our first major PCB of this course, which at its core utilizes a 555 timer circuit to drive an LED under varying output load conditions, verifying its performance and behavior is as expected as well as comparing current draw in each scenario. We also compare two different 555 IC's (NE555P and TLC555), making engineering judgments on which one of the two matches our requirements for this project better. Past driving LEDs, a requirement of our design was for the 555 to provide a $\sim 50\%$ duty cycle (with a $\pm 25\%$ margin) and a frequency of $\sim 1\text{kHz}$.

For my particular design, I decided to use one of the ones outlined in our lab manual, primarily due to its simplicity and having less components. It is also a configuration that I have used in the past and am fairly familiar with. Below is the circuit used:



This is a somewhat atypical configuration for astable operation of a 555 timer, so it's worthwhile to spend a little time here unpacking how exactly it works, in which some base knowledge of a 555 timer's operation is assumed here. The first thing that I personally noticed, was pin 7 was left floating. Pin 7 is typically the discharge pin in a 8-pin 555 timer setup, and that threw me immediately through a loop as to how this circuit actually oscillates if discharging isn't occurring. My immediate thought is that as soon as this 555 triggers, it will maintain driving its output high. However, after some digging, this isn't the case. Rather than utilizing that pin, the discharge happens through pin 3 (the output) itself, which discharges the capacitor C through R .

This circuit operates at a $\sim 50\%$ duty cycle because the same resistor R and capacitor C are used for both the charging and discharging paths. While charging, pin 3 drives HIGH, sourcing current through R into C until the voltage across C reaches $\frac{2}{3}V_{CC}$. At this point, the internal threshold comparator (pin 6) will trip, forcing our output LOW. This then allows the capacitor to discharge through R , with pin 3 now sinking our current until the voltage across C reaches $\frac{1}{3}V_{CC}$, which will trigger the comparator for pin 2 now, restarting the cycle.

Because both phases share identical component values (R and C) and traverse the same $\frac{1}{3}V_{CC}$ voltage path, the idealized RC time constant $\tau = RC$ is the same in both directions. If pin 4 behaved as an ideal switch

truly toggling between 0V and V_{CC} with zero output impedance, the charge and discharge transients would be an exact mirror of one another, thus yielding a true 50% duty cycle.

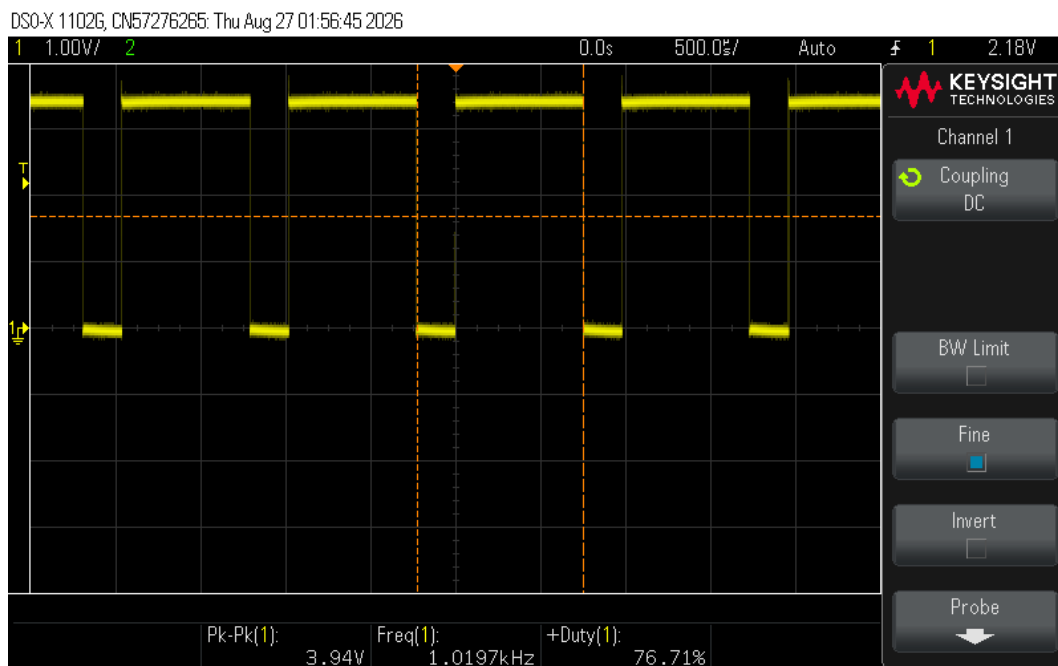
With all this background information now understood, it was time to select component values to meet my metrics. In order to achieve a 1kHz frequency at output, first chose a capacitor value of what I had on hand, which happened to be $100nF$. Thus, this determined my resistor value, which I found to be $7.22k\Omega$. As this is not a standard resistor value, I chose to "cheat" a little at first and opted for a $10k\Omega$ potentiometer as a variable resistor, only connecting the middle and one of the outside leads as opposed to all three.

Data & Results

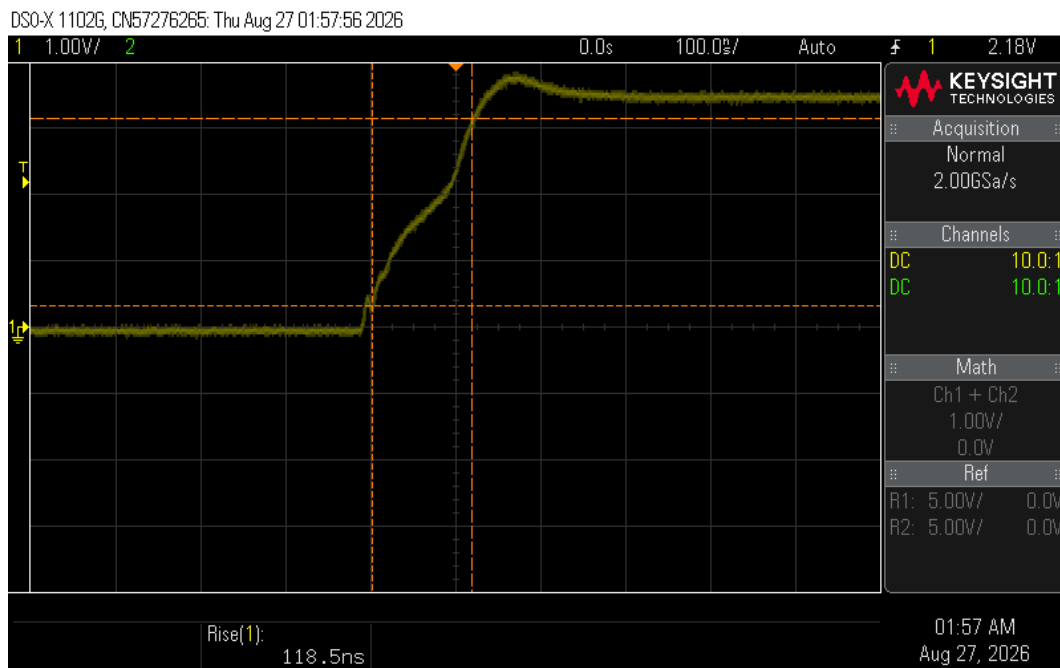
Because the fast (TLC555) and slow (NE555P) timers have very different output drive capabilities, I selected different load resistors for each device rather than using a single fixed value across both. The NE555P is rated to source $\sim 200mA$, comfortably enough to drive the LED under a $1k\Omega$ load, so I characterized it at no output load and under the $1k\Omega$ load. The TLC555, however, is only rated for $\sim 15mA$ of source current – a load much below $1k\Omega$ would demand more current than the part can actually supply. Instead, I characterized the TLC555's rise and fall time under 330Ω and $1k\Omega$ loads, both of which stay within its rated output current.

Part	Load	Rise time (nS)	Fall time (nS)
Fast 555	330Ω	18	14
	$1k\Omega$	22	15
Slow 555	No load	112	40
	$1k\Omega$	120	44

Below is an oscilloscope capture of the slow (NE555P) 555's output, showing several full cycles of the waveform on an easily readable scale. The measured frequency of $1.02kHz$ and duty cycle of 76.71% closely track the $\sim 1kHz$ frequency target, though the duty cycle deviates noticeably from the ideal 50% – this is discussed further below.



The capture below zooms in on the rising edge of the same ($1\text{ k}\Omega$) waveform, giving a measured rise time of 118.5 ns , agreeing with the 120 recorded for this configuration in the table above.



Discussion

Frequency & Duty Cycle

The measured frequency of 1.02kHz is very close to the $\sim 1\text{kHz}$ target used to select R and C , confirming the $f = 0.722/(RC)$ relationship from the datasheet held up well in practice.

The duty cycle, however, measured 76.71% against an ideal target of 50%. As noted in the Overview, a true 50% duty cycle depends on pin 3 acting as a perfectly symmetric switch, with matching effective impedances for both the charge-up and charge-down cycle. In practice, however, the output transistor itself has some hysteresis involved, and thus discharge transients do not actually mirror each other, which skeys the duty cycle away from a truly accurate 50% duty cycle. Of course for the requirements of this project, this is well within guidelines, and makes me want to continue to use this interesting configuration for the 555 timer in my final design.

It should be also noted that the TLC555's duty cycle came in much closer to the ideal 50% than the NE555p's, which is consistent for the same reasons. It helps to reiterate that the output stage of these op-amps are the causation of our unideal duty cycles, as the NE555p is a much more common bipolar totem-pole output, which causes trackable asymmetries in its pull-up and pull-down strength (which is also reflected in the rise time being $\sim 3x$ longer than its fall for the NE555p). The TLC555 however, utilizes CMOS transistors to give a much more balanced push-pull output stage, focusing the work between two different transistors who specialize in either the charge-up or charge-down phases specifically.

Load Effects on Rise/Fall Time

The load effects on specifically rise/fall time of these 555 timers are fairly clear to see, and given the explanation for varying duty cycles, is also clear to understand why these variances exist. Namely, the lower the load (i.e. the lower the resistance) added to the output, the slower the rise **and** fall times are for each circuit. This is, simply put, because the time constant goes up as you put larger resistances at the load, and less current can flow through.

The load's affect on current draw is equally understandable and simple. The lower the resistance that the output sees, the higher the current going through that part. Specifically, the LED when hooked up to the NE555p consumes roughly 2.13mA of current with a $1\text{k}\Omega$ load, and the TLC555's led draws about 2.43mA of current. The reason that the TLC can sink more current in this instance is simply due to the fact that the IC is much more capable of reaching a full 5 volts, whereas the NE555p has an offset at its output from its source voltage, thus the voltage the LED/resistor see is less in this case.

This same current-limiting behavior explains the expected LED brightness trend for the TLC555: at 330Ω the LED draws roughly 9mA (assuming a $\sim 2\text{V}$ LED forward drop against the 5V rail), versus roughly 3mA at $1\text{k}\Omega$, so the LED would be expected to run visibly brighter under the 330Ω load.

Conclusion

This lab was about designing a 555 astable circuit to drive an LED around $\sim 1\text{kHz}$ with a $\sim 50\%$ duty cycle, and figuring out which of the two timers actually fit the project better. My R/C choice nailed the frequency (1.02kHz measured), though the duty cycle landed higher than ideal (76.71%), which comes down to the asymmetric charge/discharge behavior discussed above rather than a bad component choice.

The TLC555 is clearly the faster chip by a wide margin, but that speed comes at a real cost: it can only

source $\sim 15\text{mA}$ versus the NE555P's $\sim 200\text{mA}$, so I couldn't even use the 50Ω load the lab originally called for – I had to size up to 330Ω and $1k\Omega$ just to stay within its rated output. For a project that's really just about driving an LED reliably, where edge speed doesn't matter much at these frequencies, I'd pick the NE555P. It's simpler to work with and doesn't force you to babysit your load resistor choice the way the TLC555 does.

Overall the circuit did what I expected going in, and the places where it didn't quite match the ideal math were explainable once I looked at how the 555's output stage actually behaves, not because anything was fundamentally wrong with the design.